



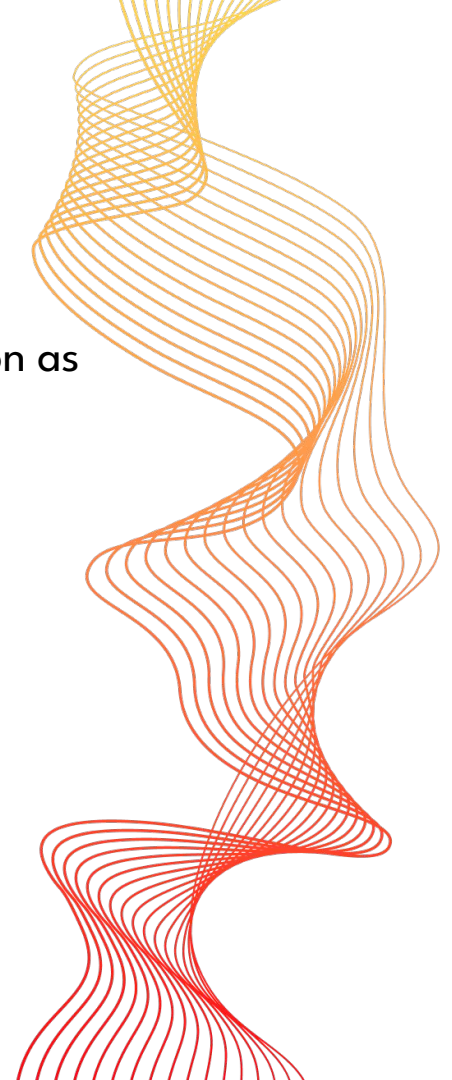
Interoperability across Various Domains in a Smart City Ecosystem using Blockchain



What is smart city

Cities made up of different systems of various domains but function as one , such as:

- Transportation systems
- Energy grids
- Waste management systems
- Public safety systems





Domains in a smart city

The smart city includes the Internet of things (IoT) devices in various domains, such as smart metering, smart grids, health care, logistics, and transport.

1. Energy:

- Smart Energy Meters: Smart meter monitor electricity, gas, and water consumption in real-time, allowing residents and utilities to track usage, detect leaks, and optimize resource allocation.

2. Healthcare:

- Remote Patient Monitoring: IoT-enabled medical devices and wearables track patients' vital signs and health metrics remotely, enabling healthcare providers to monitor and respond to changes in real-time.





Domains in a smart city

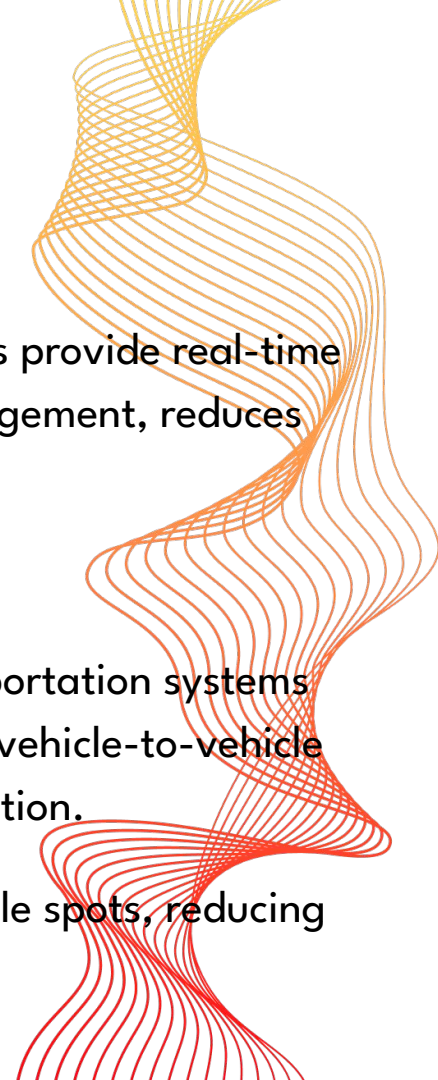
3. Logistics and Supply Chain:

- Asset Tracking: IoT sensors on goods, vehicles, and containers provide real-time visibility into the supply chain. This improves inventory management, reduces theft, and enhances route optimization.

4. Transportation:

- Connected Vehicles: IoT connectivity in cars and public transportation systems enables real-time traffic monitoring, route optimization, and vehicle-to-vehicle (V2V) communication, enhancing safety and reducing congestion.

Smart Parking: IoT sensors in parking spaces help drivers locate available spots, reducing traffic congestion and emissions caused by circling for parking.





Why Interoperability across domains is important?

- **Efficiency and Optimization:**
 - Interoperability allows efficient resource utilization.
- **Cost Savings:**
 - Interoperability can result in cost savings.
- **Disaster Management:**
 - Interconnected systems provide real-time data for emergency response, enabling effective coordination in managing disasters or crises.

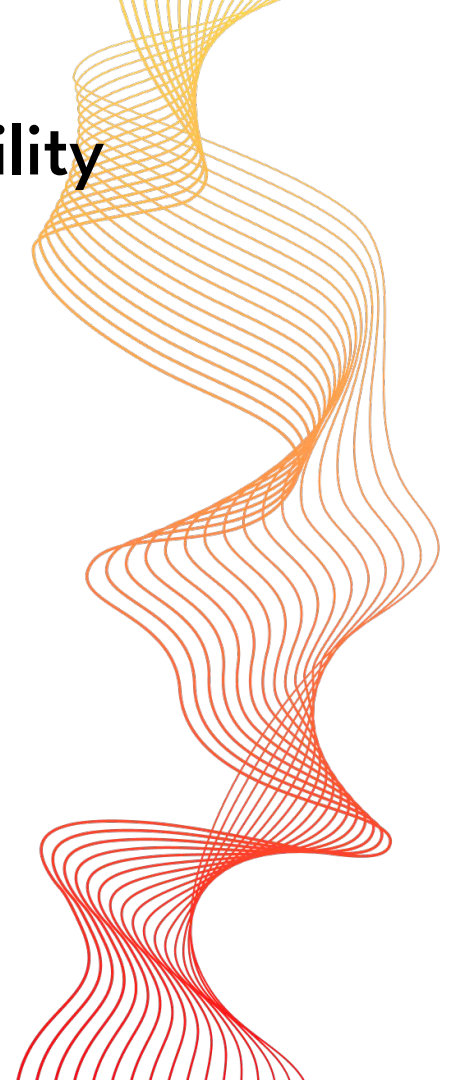
And More...





Challenges of achieving Interoperability

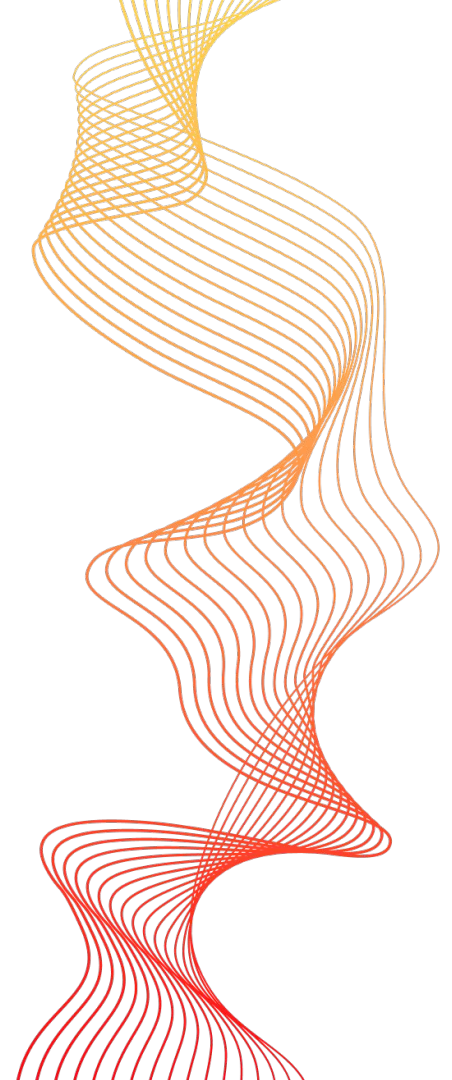
- **Heterogeneous data:** Different domains often use different data formats.
- **Privacy and Security Concerns:** Ensuring data protection while enabling data sharing is a complex challenge.
- **Data Quality:** Data quality issues, such as inaccuracies or inconsistencies in datasets, can hinder interoperability efforts.





Blockchain for Interoperability

- Decentralized digital ledger
- Immutable & transparent transactions
- Direct peer-to-peer transactions
- Diverse industry applications





Blockchain help to solve interoperability between different domains of a smart city

- **Decentralized Control**

- Blockchain's decentralized nature enables different domains to maintain control over their data while still sharing it securely.

- **Smart Contracts**

- Smart contracts can automate processes and agreements between domains. They can trigger actions when specific conditions are met.





Blockchain help to solve interoperability between different domains of a smart city

- **Data Security**

- Blockchain offers strong security features, like **decentralized storage**. This reduces the risk of data breaches and unauthorized access.

- **Data Integrity**

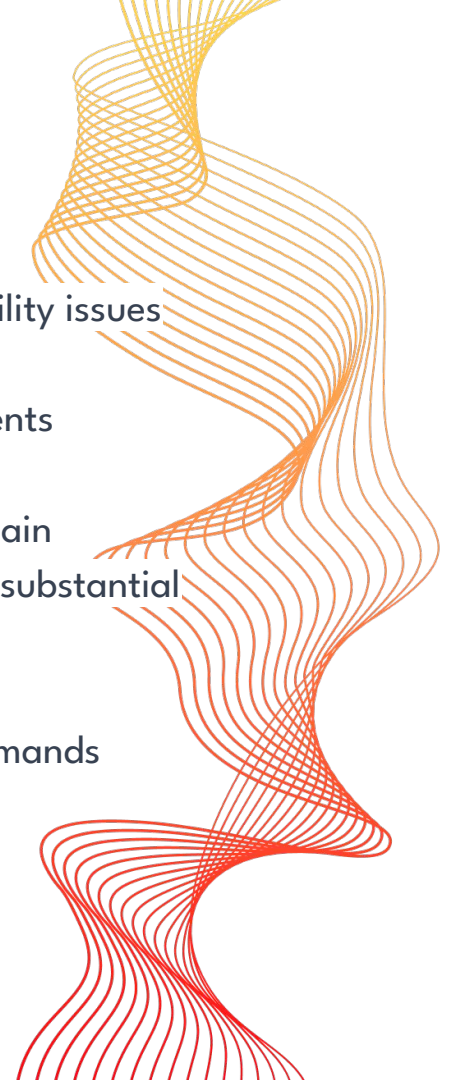
- Transactions are **tamper-resistant** and **immutable**. Hence data remains unchanged and trustworthy.





Challenges of using blockchain for interoperability

- **Scalability:**
 - Blockchain networks, such as Bitcoin and Ethereum, face scalability issues when dealing with a high volume of transactions and data.
 - Handling large volume of data exchanges for diverse requirements
- **Adoption and Integration:**
 - Encouraging stakeholders in different domains to adopt blockchain technology and integrate it into their existing systems can be a substantial hurdle.
- **Complexity:**
 - Implementing blockchain solutions is a complex process that demands significant development and integration efforts.





Developing Custom Block Chain Framework

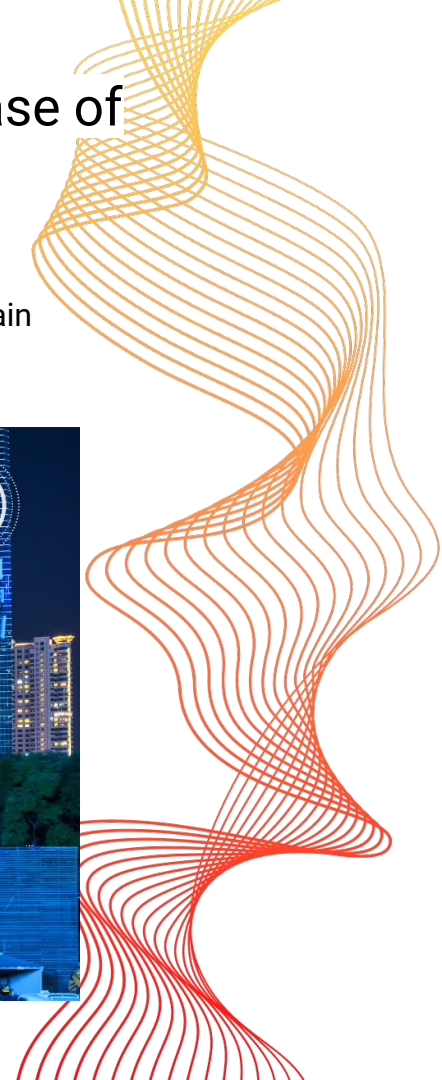
In order to work on project we would like to build our own blockchain framework or make major changes to the existing frameworks.

- **Tailored for Smart City Needs:** A custom blockchain is designed ensuring optimal performance and functionality.
- **Enhanced Scalability & Security:** Unlike off-the-shelf solutions, our custom blockchain is built to scale efficiently as the smart city project expands & unique requirements, mitigating potential vulnerabilities and protecting sensitive data , ensuring long-term sustainability.
- **Improved Interoperability:** The custom blockchain is optimized for seamless data exchange between different domains within the smart city, promoting interoperability and collaboration.
- **Hybrid UTXO and Account Model:** Our custom blockchain incorporates a hybrid model, combining the efficiency of UTXO (Unspent Transaction Output) with the flexibility of the Account model, enabling more diverse use cases.

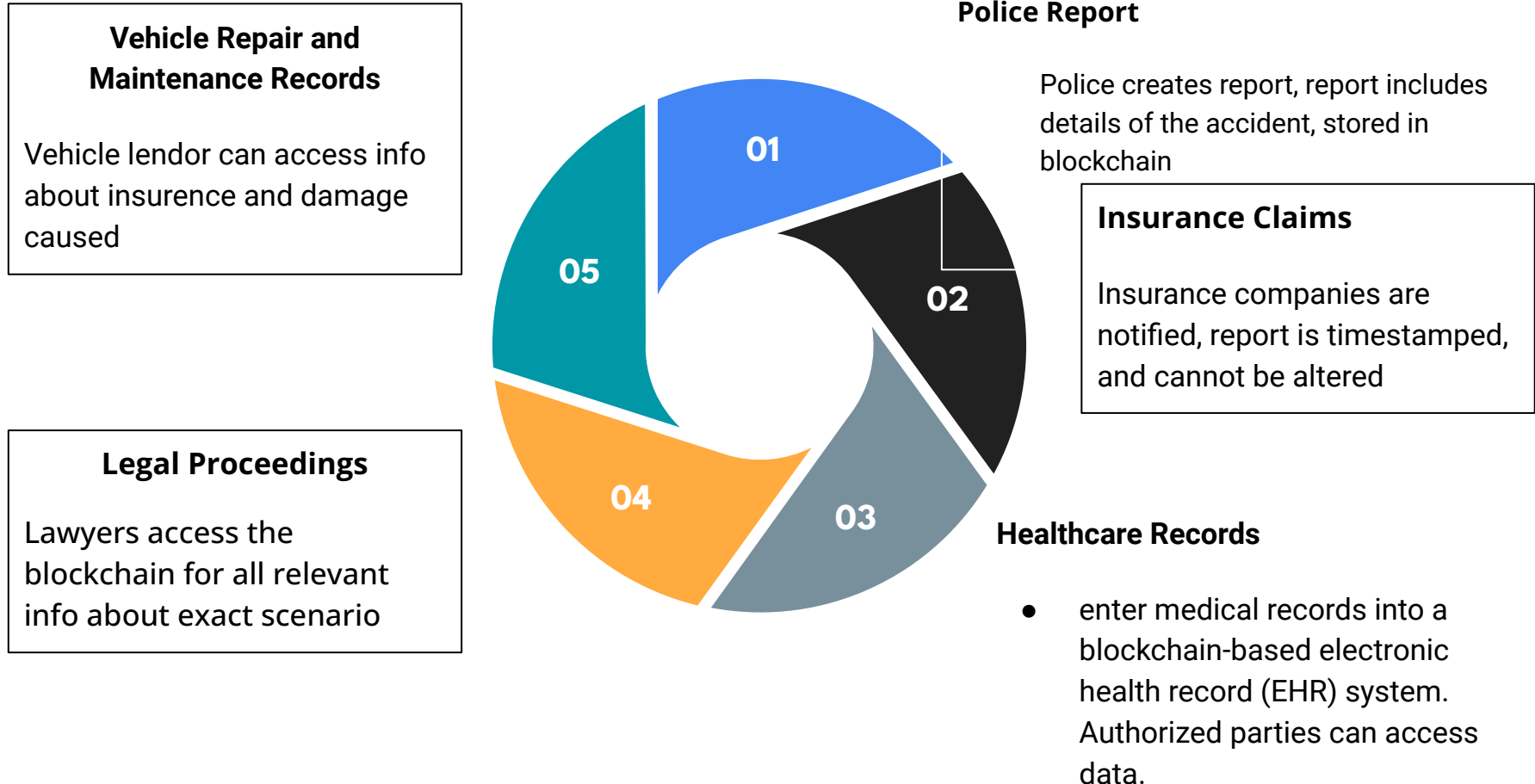


Use case: Data exchange across domains in case of traffic accident

In this scenario, we have multiple domains involved, including insurance claims, police reports, and healthcare records, and we'll show how blockchain enhances interoperability.



Use case: Data exchange across domains in case of traffic accident



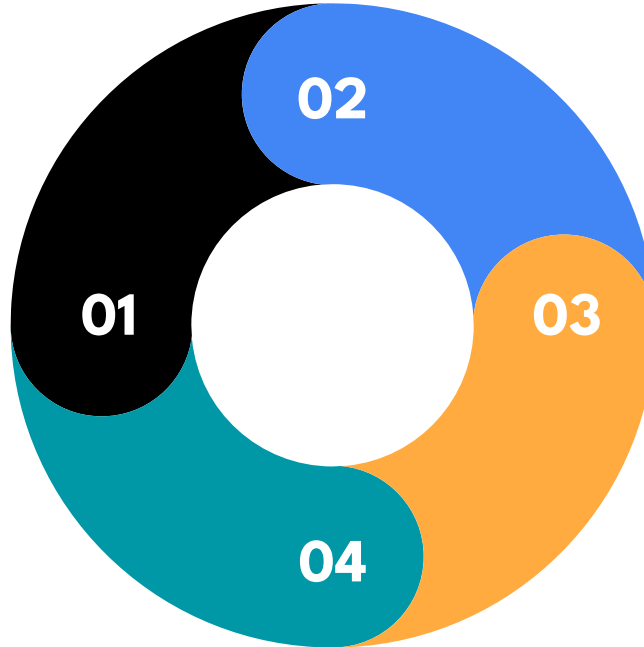
Advantages of Blockchain Interoperability for the given use case

Cross-Domain Access

Different domains can access and reference the same data on the blockchain

Security and Immutability

Data on the blockchain is secure, timestamped, and tamper-proof, ensuring the integrity of records



Automation

Smart contracts automate processes like claims processing, ensuring faster and more efficient interactions between domains.

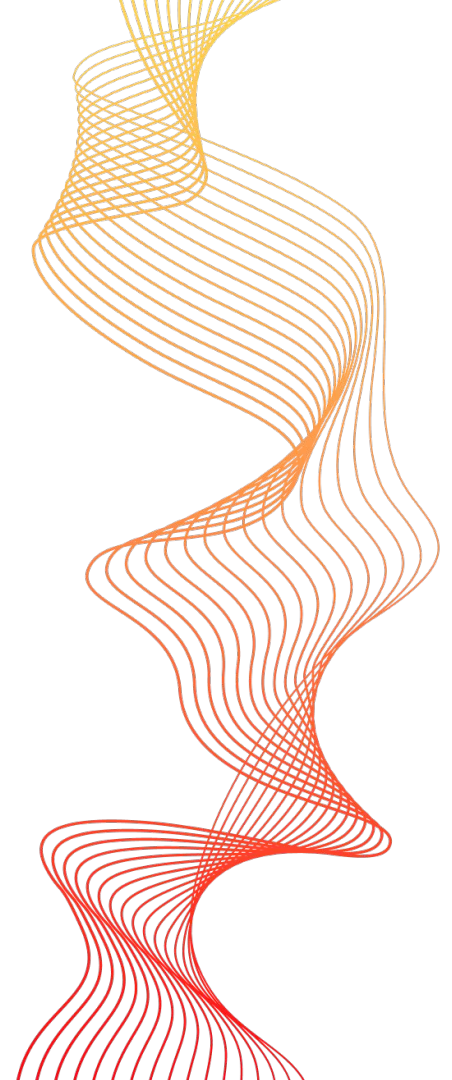
Transparency

Parties have a clear, transparent view of the entire accident-related data.



Progress Till Now

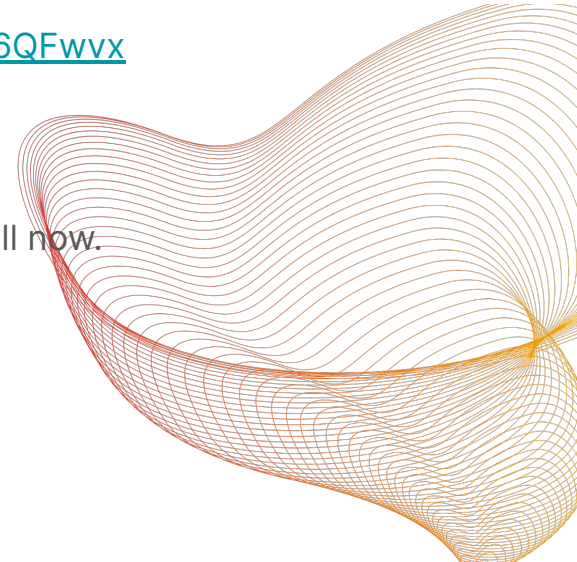
- We created a peer to peer network with a bootstrap server.
- Nodes successfully connect to its peers and messages are broadcasted across all nodes of the network .
- Duplicate messages will be deleted out, and the process of broadcasting ends when every node has got the message sent out.
- Data validation is done with help of header in the message and data can be heterogeneous as no constraint has been set on the type data i.e passed on.





Model used for Blockchain

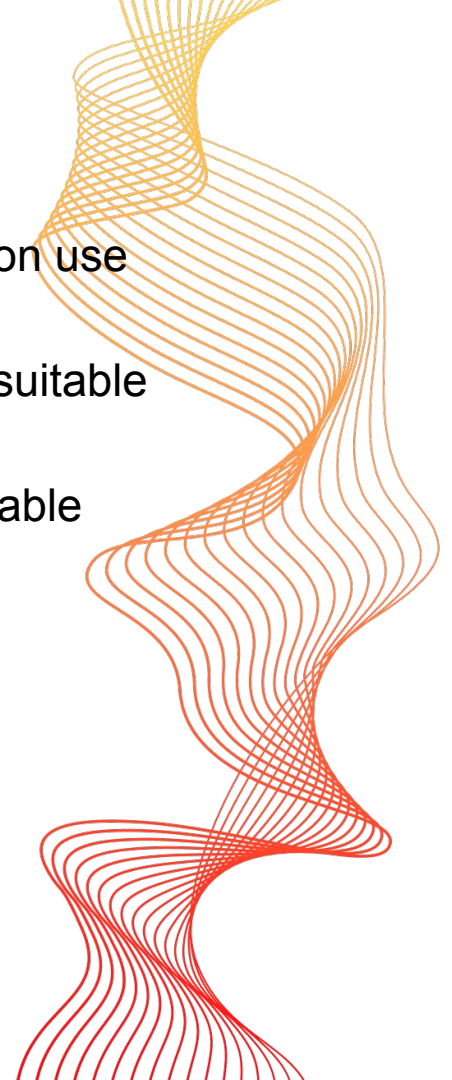
- We are currently exploring on which model is to be used for making the Blockchain robust, and make it work efficiently in real time.
- Account Model and UTXO Model are being steaded with proper maintenance of documentation at each step.
- <https://docs.google.com/document/d/1X0Y0myrYMJlU5cN6ntlvg6QFwvxYehuzgl6YaylD8/edit?usp=sharing> .
- New Blockchain technologies like SUI blockchain are also under consideration for their remarkable real time application.
- This concludes the work that has been contributed to the Project till now.





Timeline

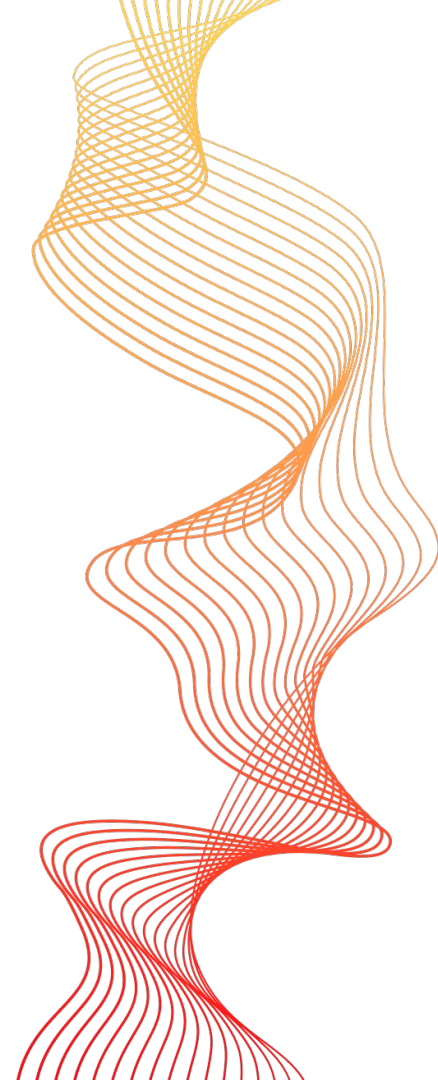
- Evaluation 1- Learned basic concepts of blockchain, worked on use case and model of blockchain
- Evaluation 2- Finalizing the framework, transaction structure suitable for data exchange across domains.
- Evaluation 3- Creating a Custom blockchain and make it scalable





References

- <https://docs.sui.io/learn/how-sui-works>
- <https://www.horizen.io/academy/utxo-vs-account-model/>
- <https://dl.acm.org/doi/pdf/10.1145/3412841.3441948>





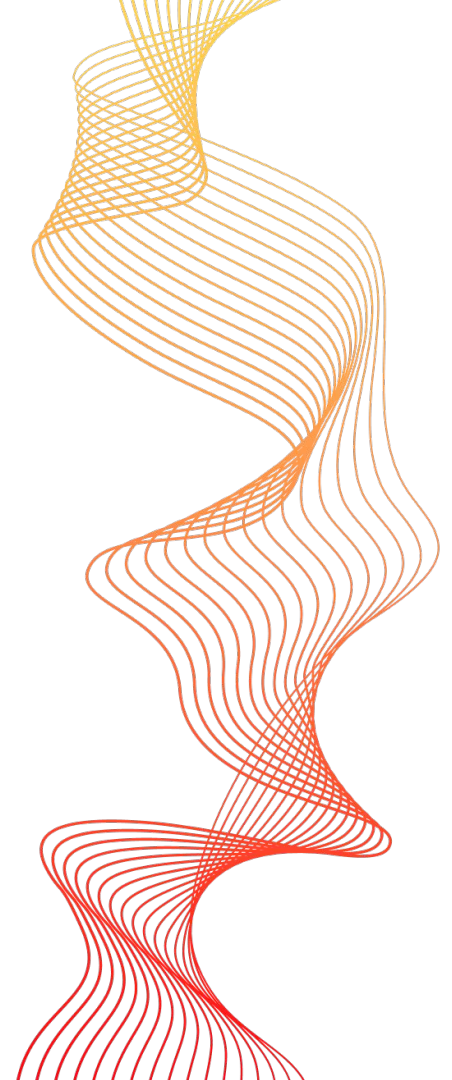
Thank you!



Traffic System

After a traffic accident occurs, the police create a digital report using a blockchain-based system. This report includes details of the accident, such as date, time, location, involved parties, and witness statements. This report is immediately stored on the blockchain.

- Data In:
 - Accident date and time
 - Accident location (GPS coordinates)
 - Vehicle information (e.g., license plate numbers, vehicle types)
 - Driver information (e.g., names, license details)
 - Witness statements
 - Initial assessment of fault or liability
- Data Out:
 - Blockchain-based Traffic Accident Report
 - Real-time accident data for traffic management and emergency services





Traffic System

Stakeholders:

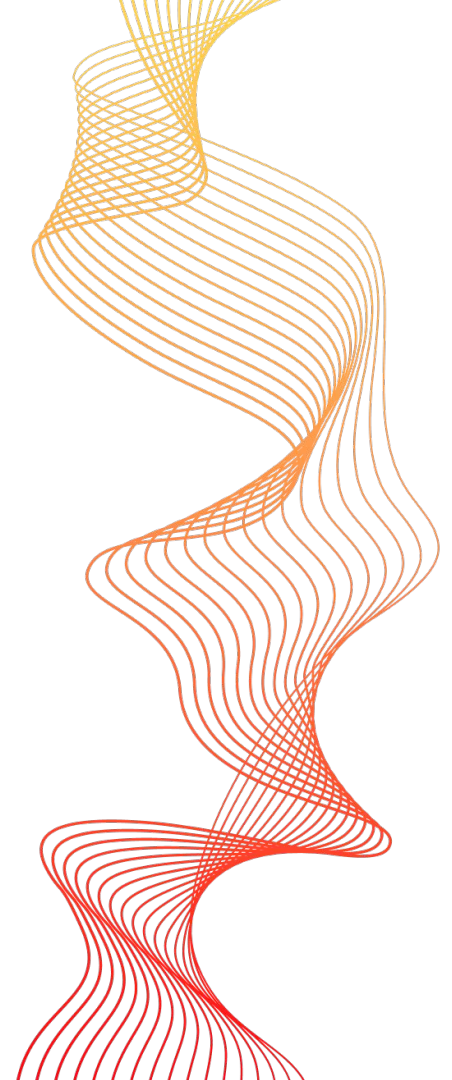
- Police departments
- Traffic management authorities
- Drivers and vehicle owners
- Witnesses to accidents

System Design:

- Blockchain-based Traffic Accident Reporting System
- Real-time data feeds from traffic management and surveillance systems
- Integration with emergency services and law enforcement databases

Authority and Hierarchy:

- Police departments have the authority to record and timestamp accident data on the blockchain.
- Traffic management authorities may have supervisory roles in managing traffic flow during and after accidents.





Traffic system

System That Is Recording:

- The Traffic System, managed by police departments, records accident data on the blockchain.

Persons with Digital Stamp:

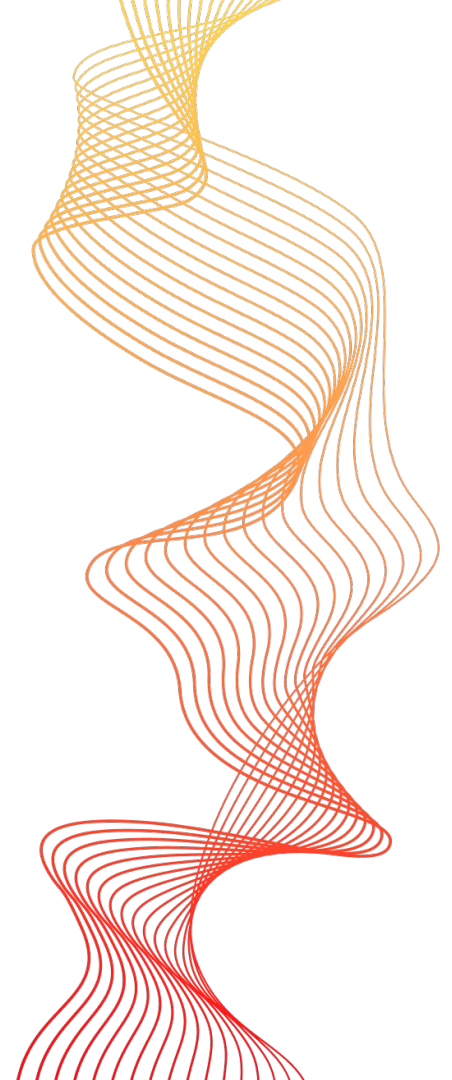
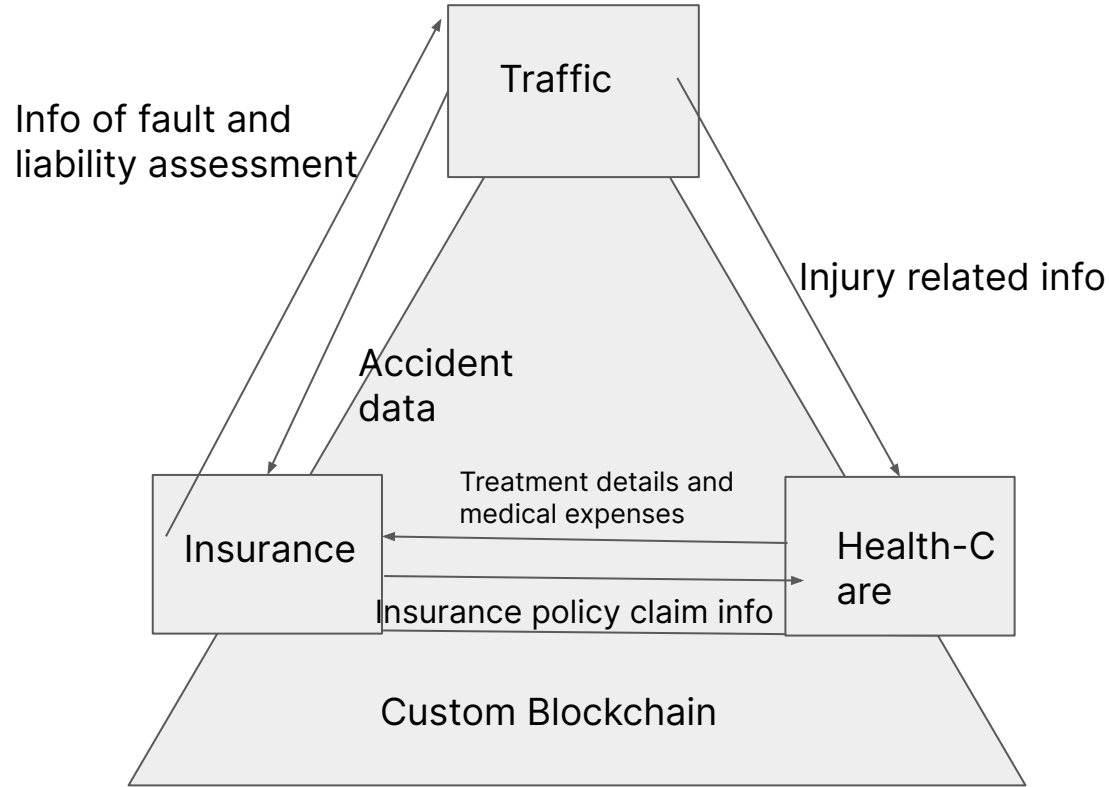
- Police officers involved in accident reporting, who have access to the system with digital signatures or authentication tokens.

Other Systems Verifying the Signature:

- Insurance companies and legal authorities may verify the digital signatures on the blockchain to ensure the authenticity of accident reports.

Permissions and Validation:

- Access to the Traffic Accident Reporting System is restricted to authorized police officers.
- Blockchain ensures the integrity of data, and digital signatures provide authenticity, preventing unauthorized alterations.





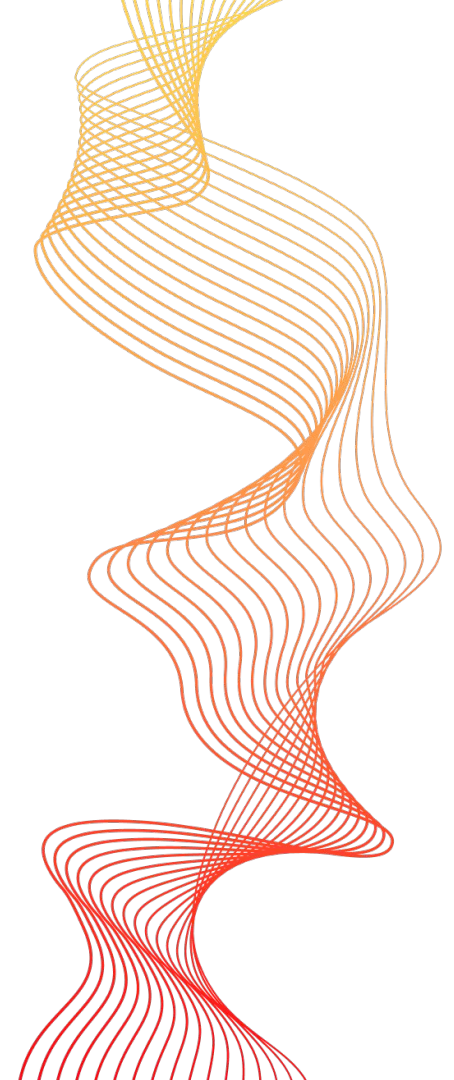
Data in: Date,time,
location,vehicle info,
driver info

Transacti
on initiate

Stakeholders:
Police departments,
Traffic management
authorities, Drivers
and vehicle owners,
Witnesses to
accidents

- Digital signature or authentication tokens
- Insurance and health care verify the signature
- Consensus
- Update the ledger
- Permission and Validation

Data out
Blockchain
based traffic
accident report





Data out
Blockchain based traffic accident report

1. Evidence collection
2. Accident details for fault assessment
3. Witness statements
4. Info of fault and liability assessment

To Insurance System-
Companies, agents

1. Date and Time
2. Location of accident
3. Names of drivers and passengers
4. Types of injuries and their severity
5. Emergency response/ medical interventions/ first aid
6. Access to Electronic Health Record: medical histories, treatment records, medical bills

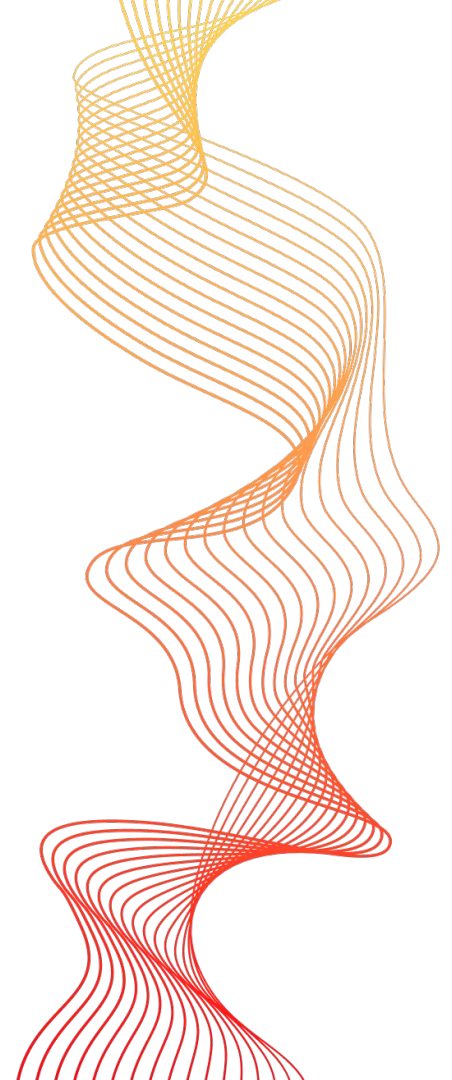
To Health Care-
Hospitals, clinics



Healthcare system

If there are injuries, healthcare providers enter medical records into a blockchain-based electronic health record (EHR) system. These records include details of injuries, treatments, and medical bills. Authorized parties, such as insurance companies and legal authorities, can access these records securely through the blockchain to verify claims related to medical expenses. Patient privacy is maintained through secure, permissioned access.

- Data In:
 - Healthcare records related to accident injuries
 - Details of treatments and medical procedures
 - Medical bills and expenses
 - Patient identification and consent (for data access)
- Data Out:
 - Electronic Health Records (EHR) on the blockchain
 - Secure access to EHR data for authorized parties (insurance companies, legal authorities)
 - Timestamped records of any changes or additions to medical data
 - Information on medical expenses for insurance claims





Healthcare System

Stakeholders:

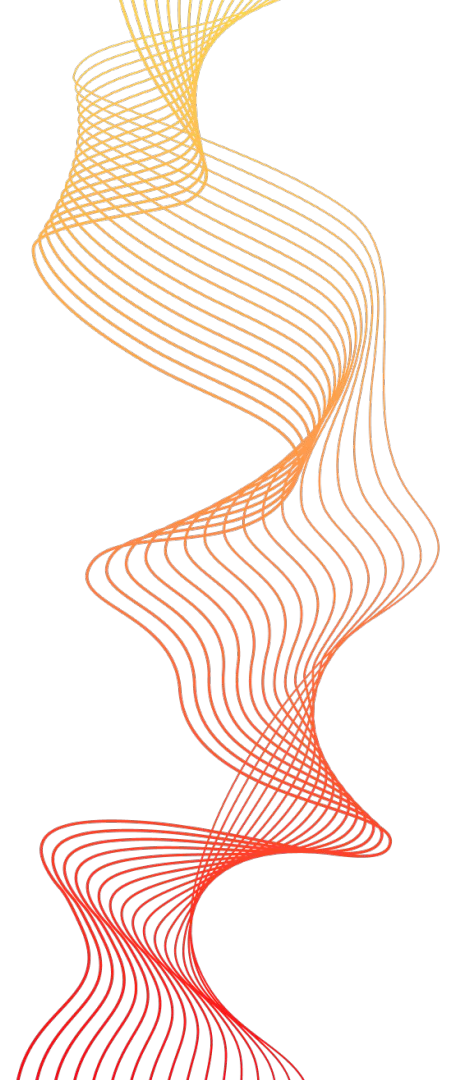
- Healthcare providers
- Patients
- Insurance companies
- Legal authorities
- Traffic System

System Design:

- Blockchain-based Electronic Health Record (EHR) system
- Secure data sharing and access controls
- Integration with insurance systems for medical expense claims

Authority and Hierarchy:

- Healthcare providers have the authority to enter and manage medical records.
- Patients have control over their health data.
- Insurance companies verify and process claims based on healthcare records.





Healthcare System

System That Is Recording:

- The Healthcare System records medical records and related data on the blockchain.

Persons with Digital Stamp:

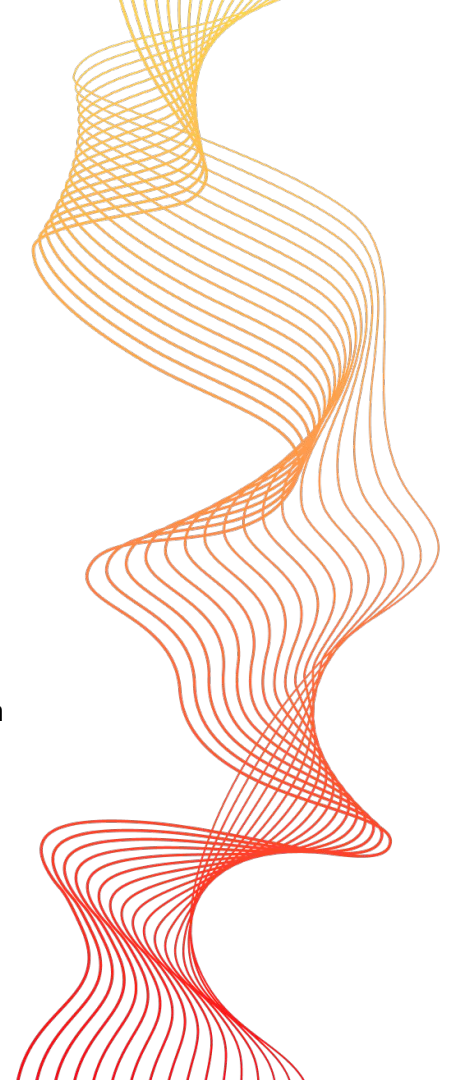
- Authorized healthcare professionals and staff with access to the EHR system.

Other Systems Verifying the Signature:

- Insurance companies and legal authorities may verify digital signatures on the blockchain to validate medical claims.

Permissions and Validation:

- Access to the EHR system is carefully controlled and limited to authorized healthcare personnel.
- Blockchain ensures the security and integrity of health records, while digital signatures verify the authenticity of data for insurance claims and legal purposes.





Data out
Blockchain based healthcare report

-
- ```
graph TD; A[Data out
Blockchain based healthcare report] --> B[1. Injury severity and status of individuals
2. Treatment status
3. Patient consent to access their health information
4. Availability for statements]; A --> C[1. Patient identification, DoB, contact etc.
2. Details of injuries, treatments, medical procedures
3. Prescription Medications
4. Medical Bills and invoices
5. Medical necessity]; B --> D[To Traffic System]; C --> E[To Insurance System];
```
1. Injury severity and status of individuals
  2. Treatment status
  3. Patient consent to access their health information
  4. Availability for statements

To Traffic System

1. Patient identification, DoB, contact etc.
2. Details of injuries, treatments, medical procedures
3. Prescription Medications
4. Medical Bills and invoices
5. Medical necessity

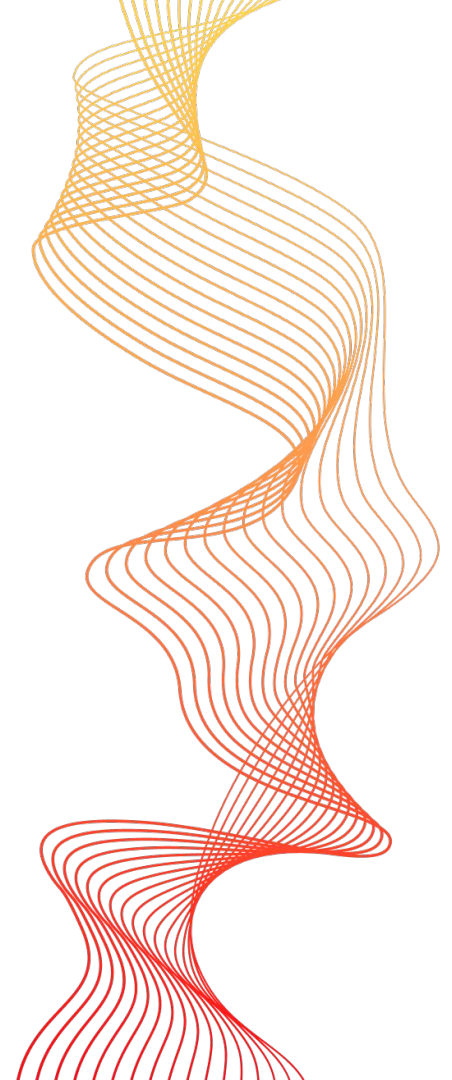
To Insurance System



# Insurance system

When the accident happens, the involved parties' insurance companies are notified. Using the blockchain, insurers can access the police report in real-time. The report is timestamped and cannot be altered, ensuring its integrity. Smart contracts are used to automate the claims process. For example, if the police report indicates that Driver A was at fault, a transaction on blockchain can trigger an automatic payout to Driver B's insurance company.

- Data In:
  - Notification of the accident
  - Access to the Traffic Accident Report on the blockchain
  - Assessment of fault and liability based on the report
  - Information related to insurance policies of involved parties
  - Claim request from policyholders
- Data Out:
  - Record of claim requests and decisions on the blockchain
  - Automatic payout transactions triggered by smart contracts
  - Claim status updates





# Insurance System

## Stakeholders:

- Insurance companies
- Policyholders
- Legal authorities
- Traffic System

## System Design:

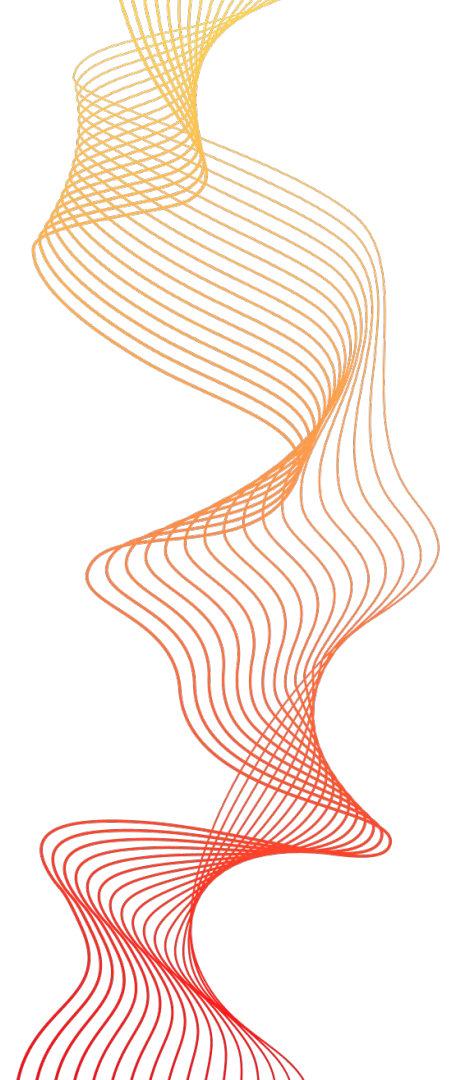
- Insurance claim processing system integrated with the blockchain
- Smart contracts for automated claims processing
- Communication with the Traffic System for accident reports

## Authority and Hierarchy:

- Insurance companies have the authority to process claims and make payments.
- Legal authorities may oversee and validate claim disputes.

## System That Is Recording:

- The Insurance System records claims and payout transactions on the blockchain.





# Insurance System

## Persons with Digital Stamp:

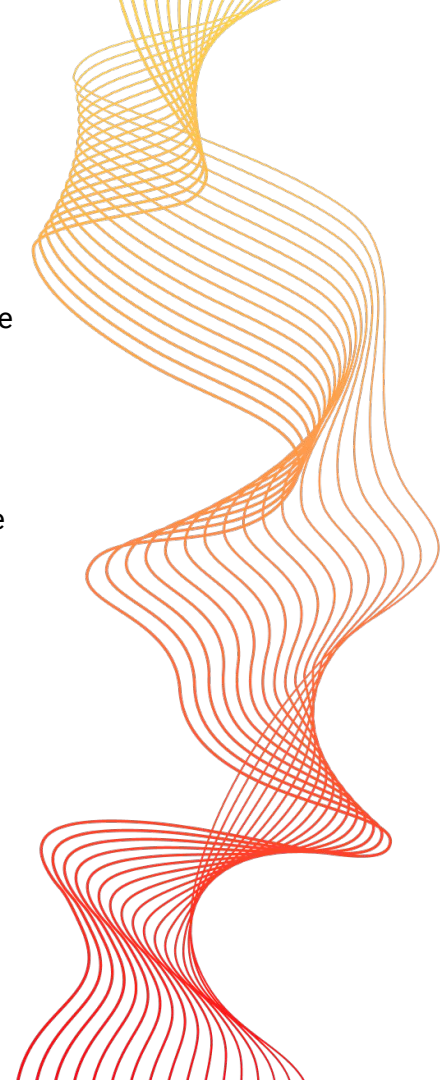
- Claims adjusters and authorized personnel within insurance companies with access to the blockchain system.

## Other Systems Verifying the Signature:

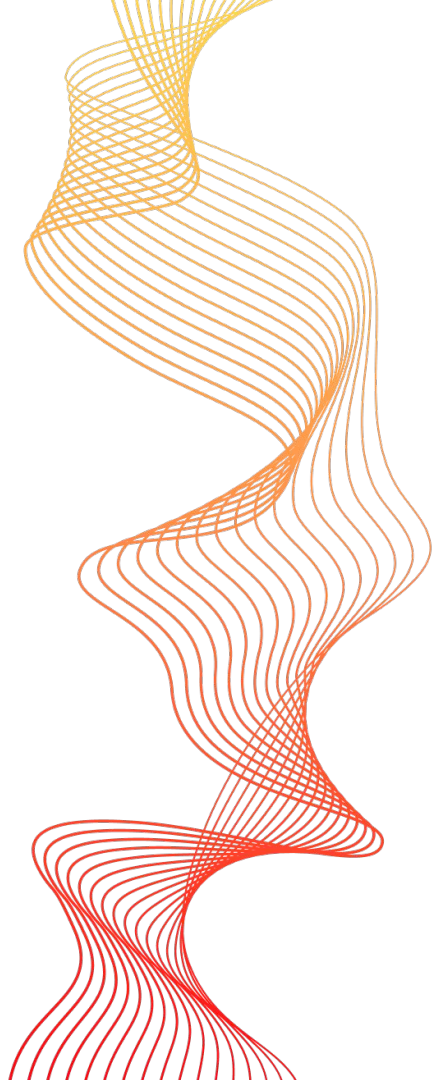
- Legal authorities may verify digital signatures to ensure claim processing and payouts are legitimate.

## Permissions and Validation:

- Access to the Insurance System is restricted to authorized personnel within insurance companies.
- Smart contracts automate the claims process based on predefined conditions and timestamps, ensuring transparency and efficiency.









Data out  
Blockchain based insurance report

1. Claim details
2. Policyholder info
3. Insured vehicle information and condition
4. Claim proofs
5. Notify about payouts and settlement modes

To Traffic System

1. Patient identification
2. Coverage details
3. Confirmation for authorisation of the patient
4. Status of request for payment procedure

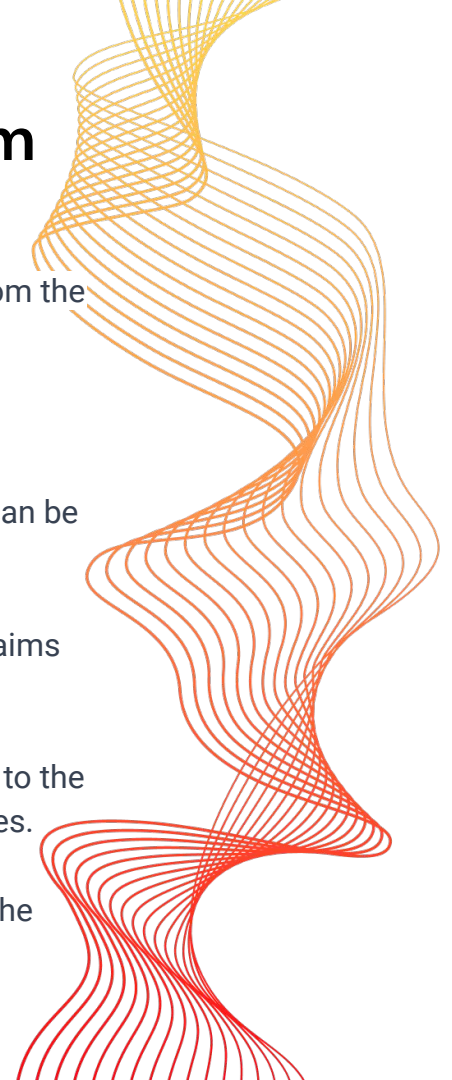
To HealthCare System





# Interoperability between those system

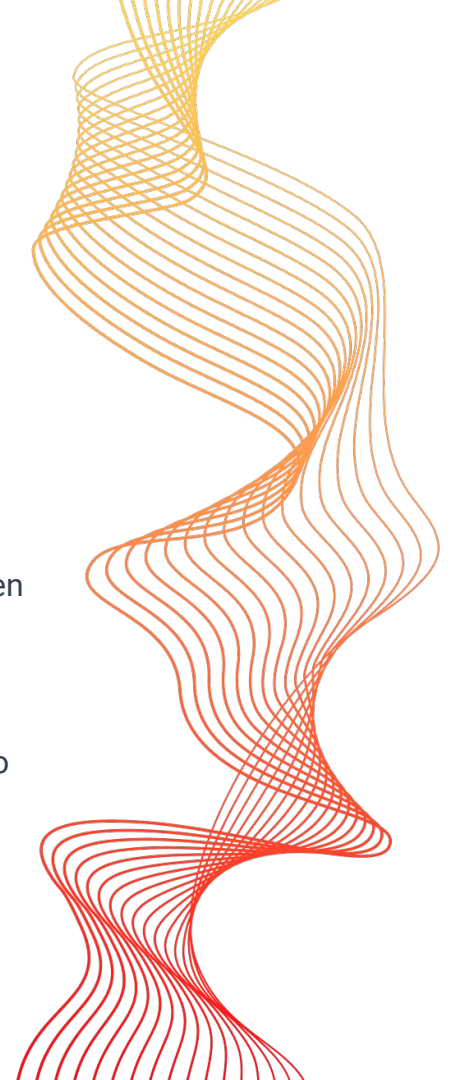
- Traffic System to Insurance System:
  - Data on the accident (time, location, parties involved, initial assessment) flows from the Traffic System to the Insurance System.
  - Insurance companies receive notifications of accidents and access the blockchain-based Traffic Accident Report.
- Traffic System to Healthcare System:
  - If there are injuries, relevant information about the accident and involved parties can be shared with the Healthcare System.
- Insurance System to Healthcare System:
  - The Insurance System may share information related to insurance policies and claims with the Healthcare System to facilitate the payment of medical bills.
- Insurance System to Traffic System:
  - Information related to fault and liability assessments can be communicated back to the Traffic System, which may be useful for accident reconstruction and legal purposes.
- Healthcare System to Insurance System:
  - Healthcare records, treatment details, and medical expenses can be shared with the Insurance System to process claims and verify medical expenses.





# Problems in this system

- privacy and security of sensitive personal and medical data
- different systems and devices can seamlessly communicate and exchange data  
ex-The Traffic System, which uses one data format, may struggle to communicate with the Healthcare System, which uses a different data format, resulting in data translation challenges.
- Data inconsistencies or inaccuracies can lead to disputes and inefficiencies.  
ex-Inaccurate information about the accident location or vehicle identification in the blockchain may lead to disputes between involved parties when determining liability.
- Gaining the trust and adoption of all stakeholders, including citizens, healthcare providers, and insurance companies, is vital for the system's success.
- Efficient coordination with emergency services and the ability to respond rapidly to accidents
- Scalability of system

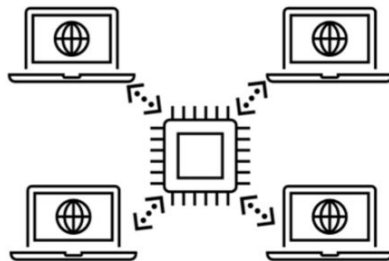




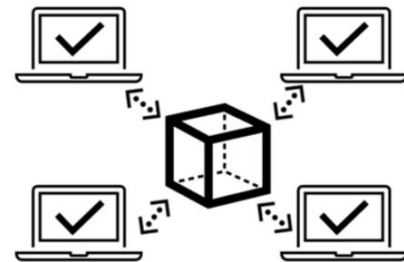
**A node raises  
transaction request**



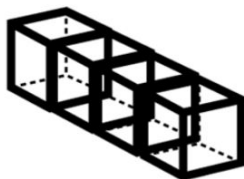
**Transaction is  
represented as a block**



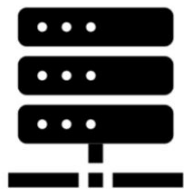
**Block is broadcasted to  
all nodes of the network**



**Transaction is validated  
by the nodes**



**A new block is added to  
the existing blockchain**



**Transaction is  
completed**

# Real Life Problems

1. Architectural Challenges: IoT devices are often connected via a centralized model known as the server/client model. All devices are identified, authenticated, and connected through cloud servers. Implementing a decentralized model, such as blockchain, requires a significant shift in the architectural design, which can be challenging
2. Security Concerns: Security is a significant concern when it comes to IoT devices. The integration of blockchain can provide a solution, but it also brings its own set of challenges. For instance, blockchain transactions require a large amount of computational power and energy, which might not be feasible for resource-constrained IoT devices
3. Interoperability: Ensuring that different blockchain networks can communicate with each other is a substantial challenge. Different blockchains have unique block structures, consensus mechanisms, smart contracts, and authentication mechanisms, making it difficult to establish a standard for interoperability
4. Data Privacy: IoT devices collect a massive amount of data, and ensuring the privacy of this data is crucial. While blockchain can provide a solution, achieving privacy-preserving data sharing in a decentralized system is a significant challenge
5. Scalability and Throughput: Blockchain networks typically face scalability issues due to the large number of transactions they need to process. This becomes even more challenging when IoT devices are involved, as they generate a vast amount of data that needs to be processed in real-time
6. Legal and Regulatory Issues: Blockchain technology still faces numerous policy and legal questions. The lack of clear regulations and policies associated with digital or cryptocurrencies, as well as data ownership and access, are significant hurdles for the adoption of blockchain in IoT



## FYP EVALUATION-2

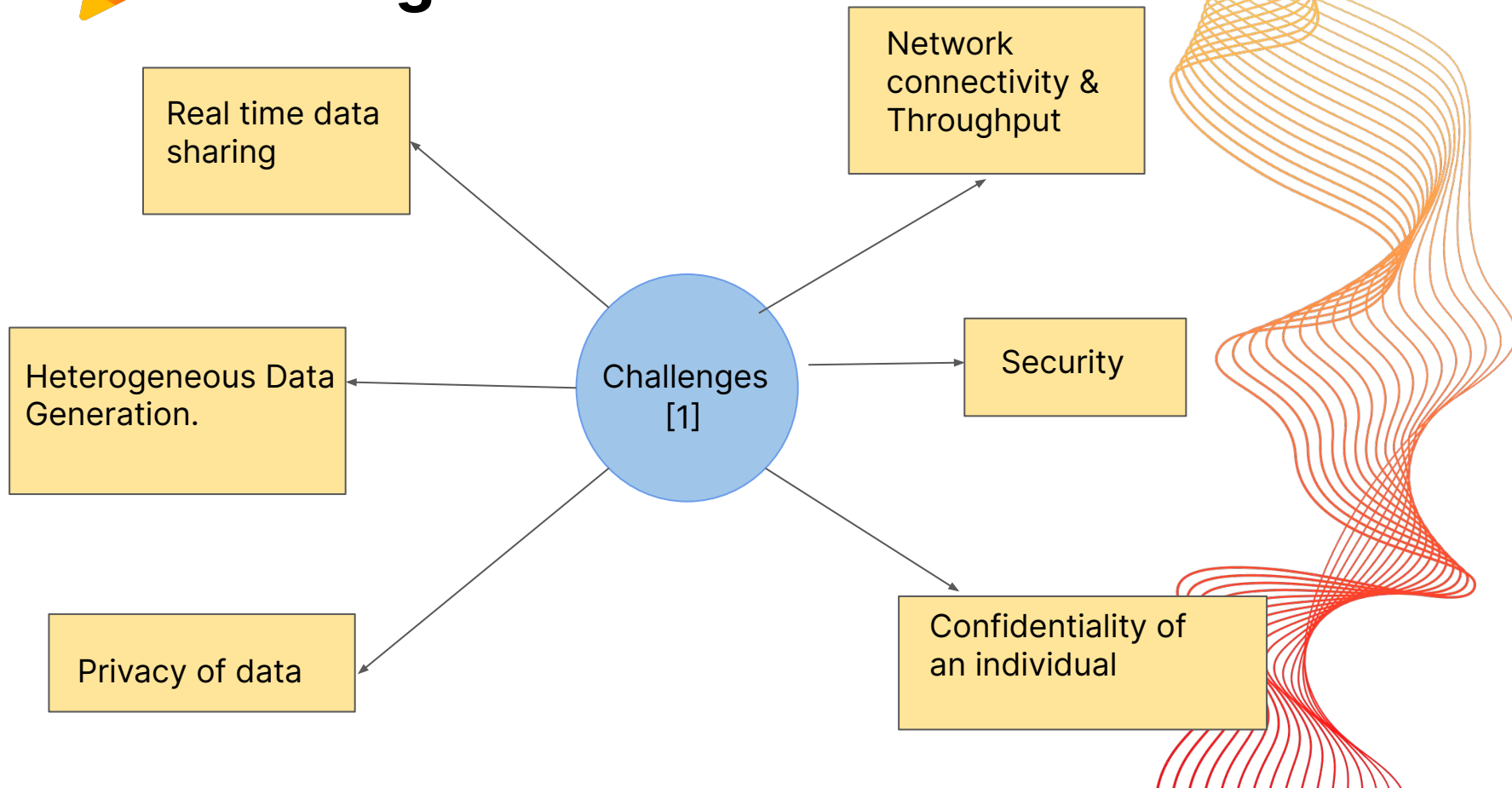


# Challenges

- With IoT, a large amount of data is being captured from all around people by millions of devices.
- So there are reasonable concerns with respect to scalability, reliability and security, when considering the creation of large IoT networks.
- So it is essential to provide for an acceptable level of confidence in the platforms that will power the Internet of Things.
- Besides, a key question is if optimum economic value could be derived from the vast amount of data generated.
- Globally, only about 30% of IoT projects survive beyond the pilot phase.
- This is a powerful indicator that much needs to be done to effectively tap this new technology.



# Challenges



# Privacy



**Ledger is shared, but participants require privacy**



**Participants need: – Transactions to be private – Identity not linked to a transaction**

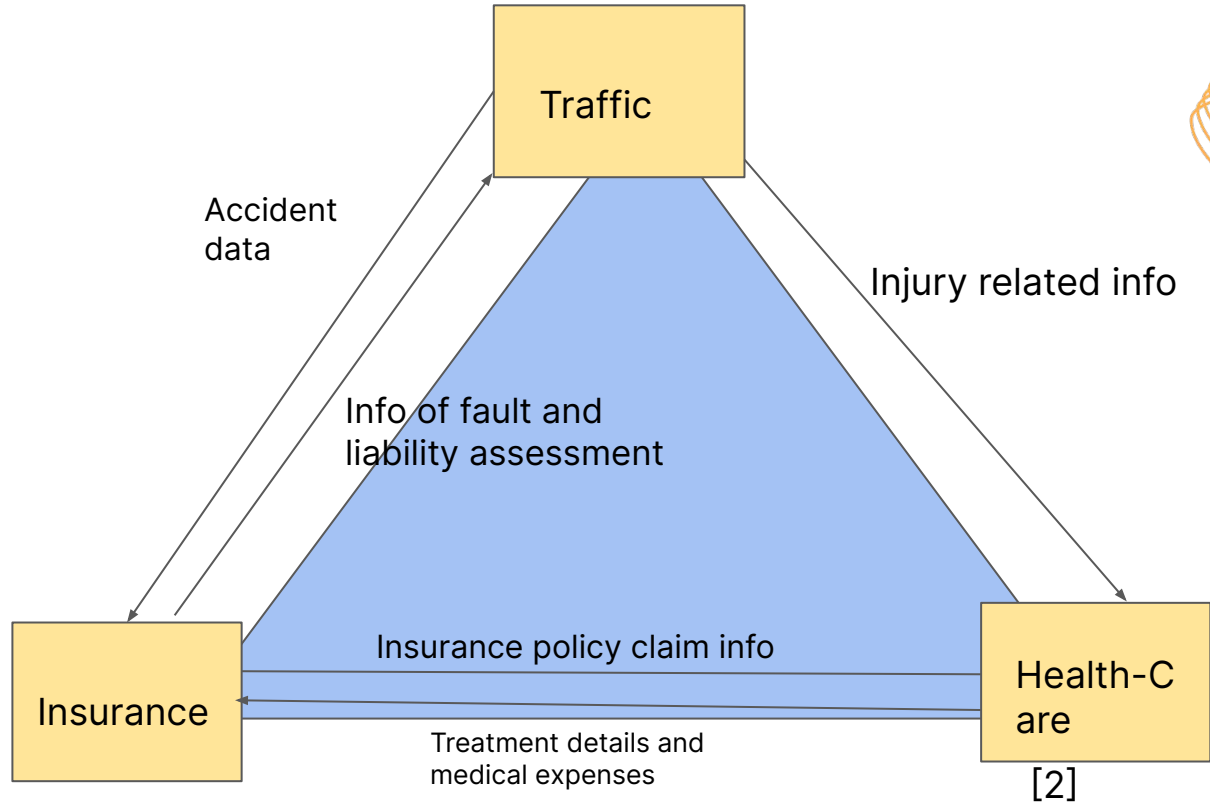


**Transactions need to be authenticated**

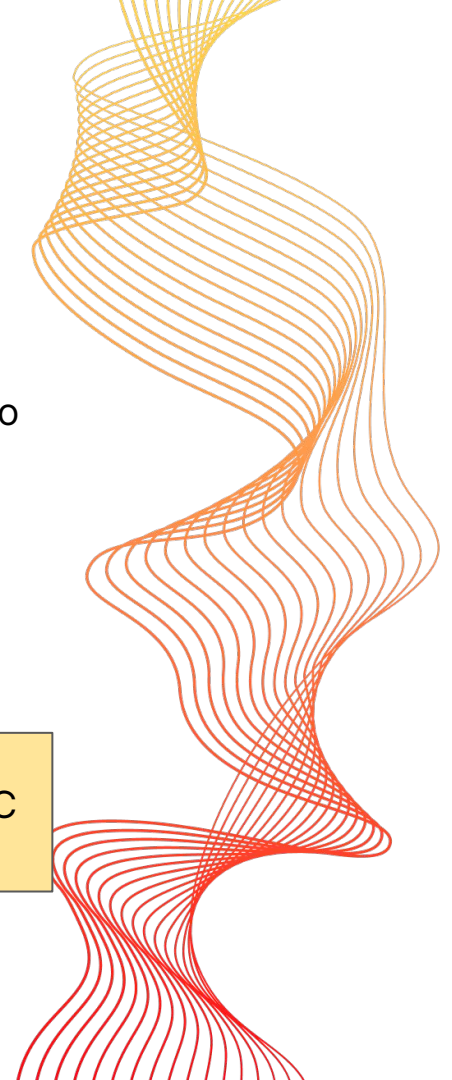


**Cryptography central to these processes**

# Systems in Blockchain



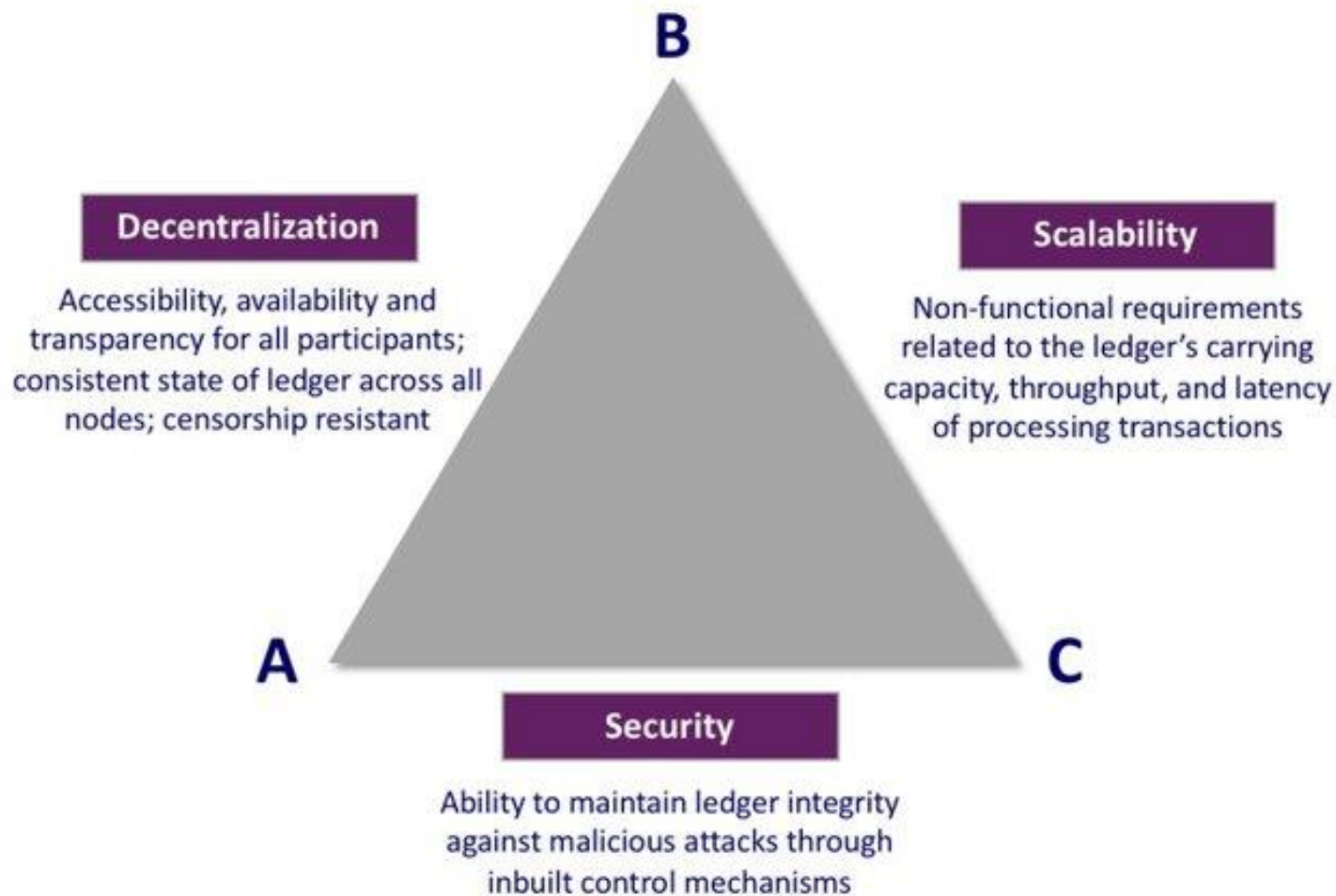
Custom Blockchain





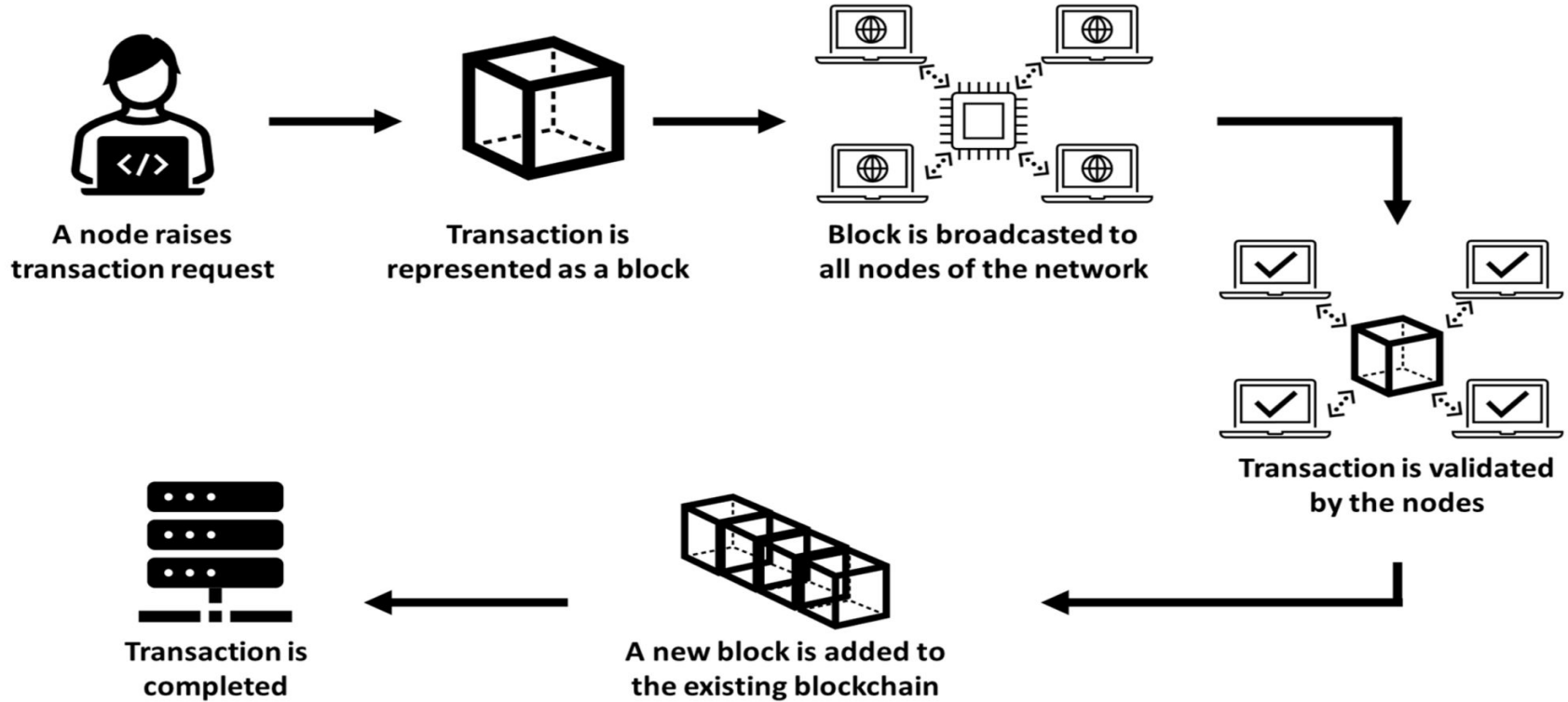
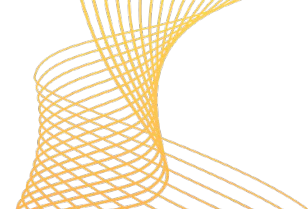
# Scalability & Throughput

- Blockchain networks are designed to be decentralized and secure, which often results in a trade-off with scalability.
- In a traditional blockchain, every node in the network must validate and store all transactions. As more transactions are added, the network can become slower and less efficient.
- Throughput refers to the number of transactions a blockchain can process within a given time frame.
- Many blockchain networks struggle with slow confirmation times and low throughput, which can be a significant issue when dealing with real-time IoT data.





# Flow Diagram





# Progress in the Implementation

## Messaging Security:

- Introducing RSA encryption methods and digital signatures to ensure message integrity, authenticity, and security in peer-to-peer communication, setting new standards for secure messaging frameworks.

## Custom Blockchain Protocol:

- Developing a tailored protocol for efficient message traversal within blockchain systems, optimizing transactions and data flow, enhancing the overall functionality of blockchain networks.

## Serialization with JSON :

- Utilizing JSON as an interim solution for effective data serialization, paving the way for future innovations in serialization techniques, ensuring streamlined data processing and transmission.

## Future-Ready Decentralization:

- Enabling seamless integration with diverse applications and platforms, emphasizing adaptability and scalability, making the framework future-ready for evolving demands in the decentralized communication landscape.

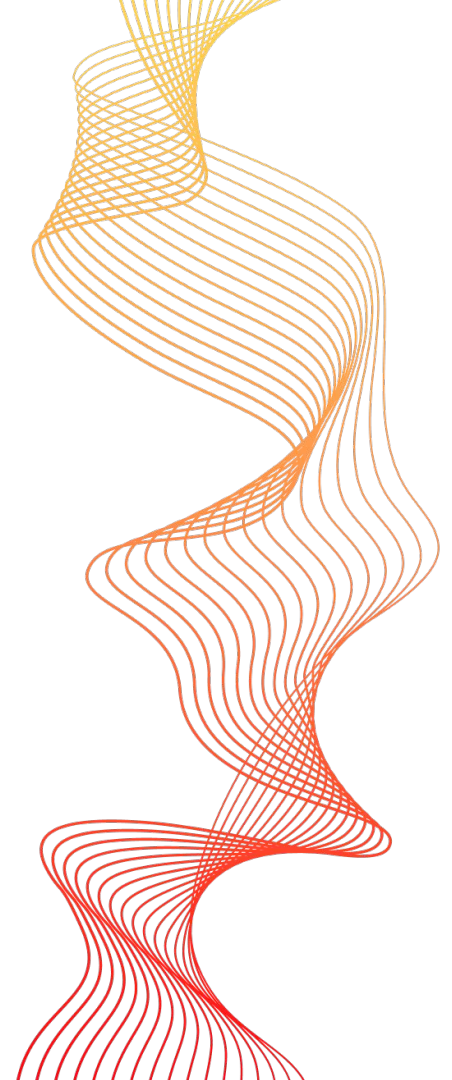
## Transformative Impact:

- Showcasing the framework's potential to reshape digital communication paradigms, fostering collaboration, innovation, and secure data exchange, ushering in a new era of interconnected and efficient peer-to-peer networks.



# References

1. Reegu, F., Daud, S. M., & Alam, S. (2021). Interoperability challenges in healthcare blockchain system-A systematic review. *Annals of the Romanian Society for Cell Biology*, 15487-15499.
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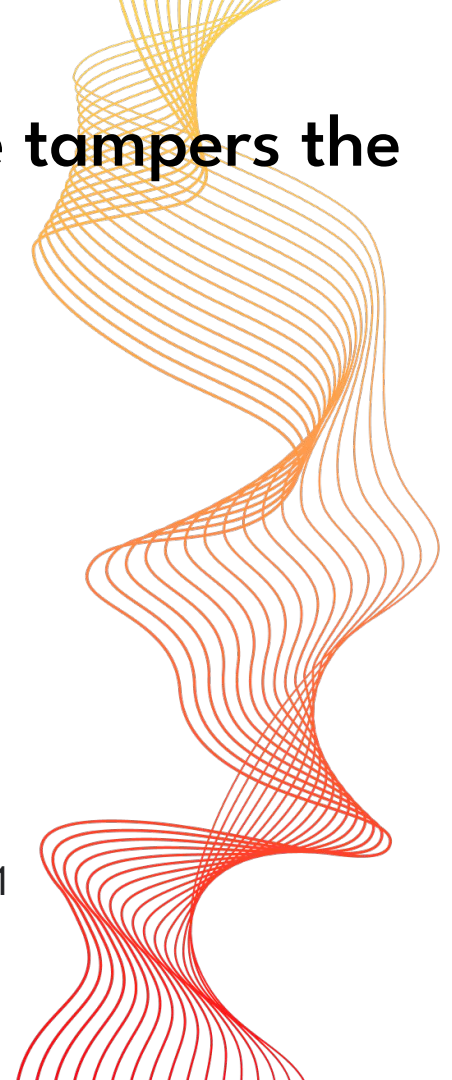




## Question : what happens if someone tampers the block?

Ans : If a hacker tries to tamper with a block, the hash of the block changes, hence changes the hash of the subsequent blocks. To propagate a change across the blockchain, 51% of the network would have to agree to it which is next to impossible. That's how blockchain is amazingly reliable and innovative.

Number of meets conducted with Faculty Dr PVN Prasanth : 11





# FYP FINAL EVALUATION



1.

